

Project Context: FlashShield (SIH 2026)

1. Problem Statement & Background

- Problem Statement ID:** 26192
- Title:** Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme
- Organization:** Ministry of Home Affairs (MHA)
- Department:** National Disaster Response Force (NDRF), Disaster Management Division
- Category & Theme:** Software | Disaster Management

The Ground Reality

Hilly Himalayan catchments are prone to cloudbursts and sudden flash floods with warning times often under an hour. Traditional weather forecasting relies on broad, district-level radar and satellite models that fail to predict localized, high-velocity surges funneling down narrow river valleys. Upstream flash floods frequently submerge settlements and wash out roads before local disaster authorities can react.

FlashShield bridges this gap by merging digital elevation terrain physics, real-time river telemetry rates of rise, and automated tactical evacuation routing for field response units.

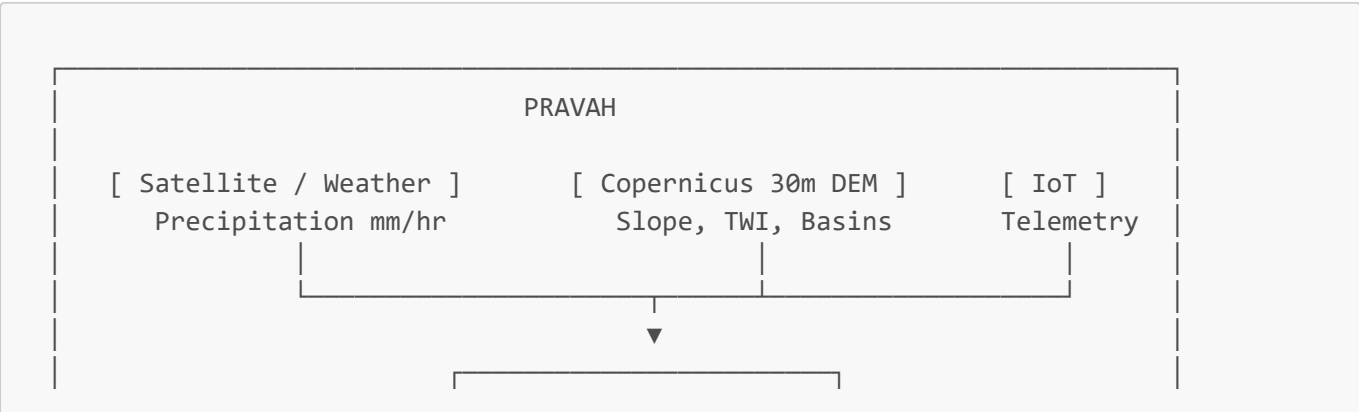
2. Study Area Anchor: Mandakini River Valley

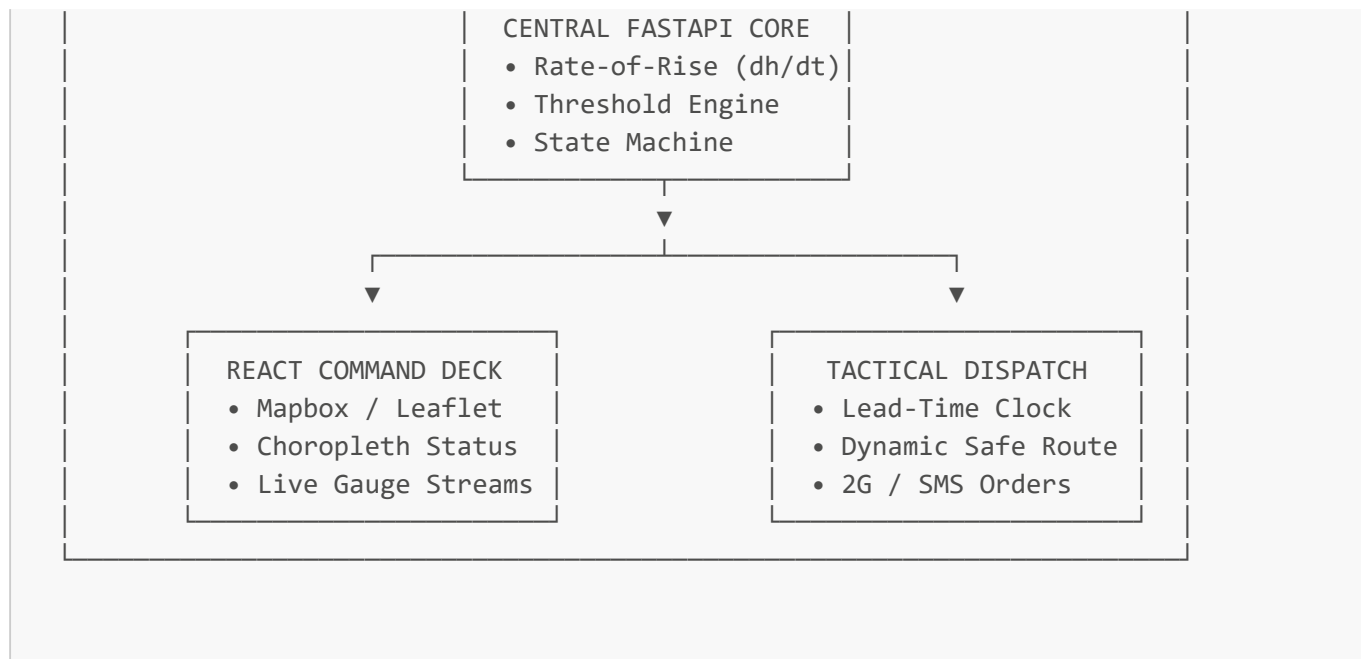
To make the system realistic, concrete, and scientifically defensible, the prototype is strictly bounded to the **Mandakini River Basin (Rudraprayag District, Uttarakhand)** around coordinates **30.28° N, 78.98° E**.

Target Settlements Monitored

- Tilwara (VIL_TILWARA):** High-vulnerability settlement situated at a low river-bend contour (Population: ~1,840).
- Sumerpur (VIL_SUMERPUR):** Intermediate valley settlement (Population: ~920).
- Rudraprayag Town (VIL_RUDRAPRAYAG):** Downstream junction confluence (Population: ~5,400).

3. High-Level System Architecture





4. The Two-Stage Roadmap

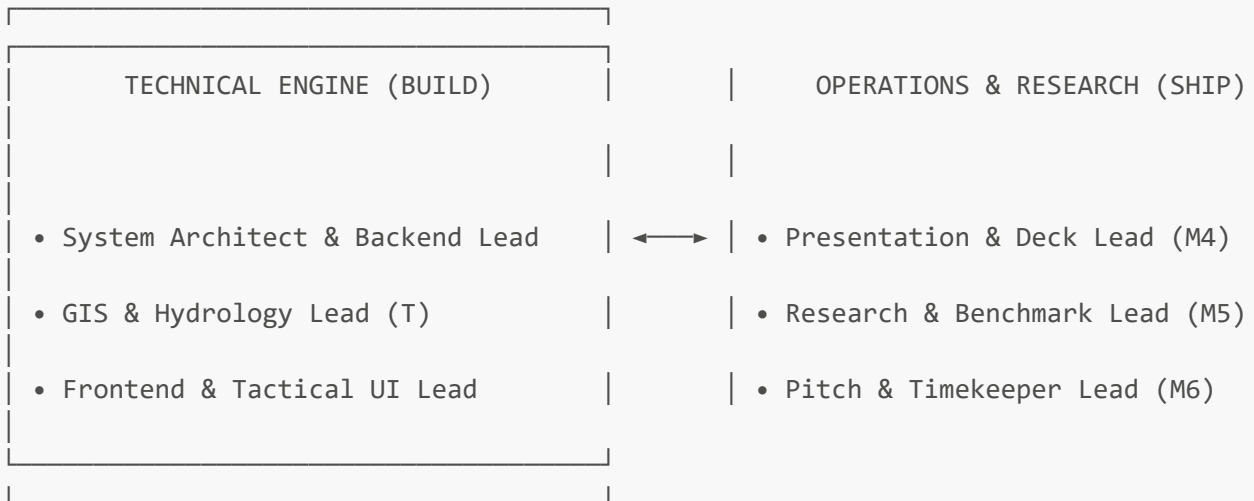
Stage 1: Prototype Demo (Deadline: September 10, 2026)

- **Goal:** Clear the internal college shortlisting round with a working end-to-end prototype.
- **Scope:**
 - Interactive dark-mode GIS map loaded with real Mandakini village polygons and river vectors.
 - Simulated IoT telemetry streaming live via WebSockets.
 - On-demand cloudburst surge injection triggering hydrodynamic rate-of-rise (dh/dt) thresholds.
 - Instant state transition of Tilwara from **NORMAL** (Green) to **CRITICAL** (Red).
 - Automated pop-up of the **NDRF Incident Order Modal** displaying a calculated 42-minute lead window and an uphill evacuation route to the designated shelter.

Stage 2: Full System Expansion (Deadline: September 17–20, 2026)

- **Goal:** Finalize the production codebase and submit the official 6-slide PDF deck to the SIH portal.
- **Scope:**
 - Topographic Wetness Index (TWI) and drainage matrix generated from raw 30m DEM rasters.
 - Pre-trained XGBoost 6–24h macro-vulnerability classifier.
 - SHAP explainability factor breakdown ("Why this risk?").
 - Dynamic, risk-weighted Dijkstra evacuation routing using OpenStreetMap road network graphs.
 - Hardware telemetry anomaly/sanity filters (stuck-value, impossible jumps, heartbeat watchdogs).

5. Team Structure & Work Distribution



- **System Architect & Backend Lead:** Owns FastAPI core, WebSocket feeds, threshold logic, API contract enforcement, and module integration.
- **GIS & Hydrology Lead (T):** Owns study area polygons, river geometries, shelter points, DEM topographic calculations, and road graphs.
- **Frontend & UI Lead:** Owns the React/Tailwind Command Deck, Mapbox/Leaflet rendering, real-time metric cards, and Incident Modal.
- **Presentation & Compliance Lead (Member 4):** Owns formatting, strict alignment to the official SIH 6-slide template, and PDF conversion.
- **Research & Domain Lead (Member 5):** Gathers real geographic elevation numbers, historical 2013 flood telemetry reference marks, and incident text.
- **Pitch & Rehearsal Lead (Member 6):** Scripts the 90-second live demonstration, times pitch runs, and coordinates panel defense.

6. Standardized Development Principle: Mock → Replace

No team member waits for another to finish their component before starting:

1. **Frontend** designs against `src/mockData.js` and dummy GeoJSON vectors from Day 1.
2. **GIS Lead** draws study polygons and exports standard GeoJSON files (`gis/outputs/`) directly consumable by Frontend and Backend.
3. **Backend Lead** provides mock JSON endpoints matching `docs/API.md` before wiring real business logic.
4. All three technical layers continuously integrate and replace mock layers without breaking interfaces.

7. The 90-Second Demo Script (September 10 Presentation)

1. **0:00 – 0:20 (The Problem):** Present the core limitation of existing systems—valley settlements have zero warning when cloudbursts happen miles upstream.
2. **0:20 – 0:40 (Baseline Operation):** Showcase the live Command Deck. Gauges show normal rainfall (14 mm/hr), river level (110 cm), and green village polygons.
3. **0:40 – 1:00 (The Surge Event):** Trigger `POST /api/simulate/cloudburst`. Upstream gauge rate of rise accelerates (+0.4 → +1.2 → +3.8 cm/min).
4. **1:00 – 1:20 (Actionable Response):** Threshold engine trips $dh/dt > 2.0$ cm/min. Tilwara instantly turns **RED**. The Incident Modal pops up displaying:
 - *Lead Time:* ~ 42 minutes to peak crest.
 - *Muster Shelter:* High School Grounds via uphill corridor.
 - *Action:* Sound valley alarm horn, deploy field unit.
5. **1:20 – 1:30 (Roadmap Defense):** Conclude by showing the post-shortlist expansion plan (integrating DEM physics, SHAP values, and risk-weighted graph routing) for the final SIH submission.